SWAMI KESHVANAND INSTITUTE OF TECHNOLOGY, MANAGEMENT & GRAMOTHAN, JAIPUR

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MAJOR PROJECT 50% CHECKPOINT ARCHITECTURE SUMMARY REPORT (FORM-3)

Project Title: Smart Learning Planner | **Evaluation Date:** October 31, 2026 | **Academic Session:** 2026–2027

---

1. Project Team & Module Ownership

Roll No.	Student Name	Assigned Project Module	Key Milestone Deliverables
23SKCS022	**Himanshu Jain**	AI Prompts, Schemas & Diagnostics	Pydantic Schema, Rule #9, WeakTopicsService, Isolation Tests
23SKCS001	**Devanshu Jangid**	Frontend UI & Full-Stack Integration	Next.js Dashboard, TaskExecutionModal, E2E Test Suite
23SKCS015	**Khushi Mathur**	Recommendation Engine Pipeline	TF-IDF Vector Space, Coursera Catalog (3,522 items), Cosine Sim
23SKCS019	**Heman Parashar**	Backend Services & Database	FastAPI, Clerk JWT Auth, Supabase RLS, Plan Versioning

2. Deterministic Data Flow & Zero-Hallucination Guardrails

To protect academic evaluations against generative AI hallucinations, the system implements a strict 4-phase deterministic loop: (1) **Structured Scheduling** via Gemini bounded by Pydantic schemas; (2) **Diagnostic Telemetry** using mathematical formulas without black-box AI claims; (3) **Grounded Recommendations** mapped into a 3,522-course TF-IDF vector space; and (4) **Adaptive Re-allocation** guaranteeing daily study hour invariants.

Execution Stage	Component	Deterministic Guardrail & Implementation
Phase 1: Ingestion	Gemini 1.5/2.0 Flash + Pydantic	Rule #9 forces 3-8 domain keywords combining subject + skills; Pydantic enforces chronological days & daily hours ceiling.
Phase 2: Diagnostics	WeakTopicsService	Strict empirical formulas (Study duration ratio $\rho < 0.50$, deficit $\Delta \geq 15$m, missed sessions). Zero AI guesses.
Phase 3: Retrieval	TF-IDF Vector Space	Cosine similarity over 3,522 verified Coursera courses with $\tau \geq 0.10$ cutoff; prevents phantom or hallucinated URLs.
Phase 4: Adaptation	AdaptivePlanInput + 10 Invariants	Forward-only allocation ($d \geq$ current_day), strict daily hours ceiling (compresses low-priority sessions), audit versioning.

3. Mathematical Diagnostic Formulas (Weak Topic Detection)

Metric / Rule	Mathematical Formulation	Diagnostic Outcome & Severity
Study Duration Ratio	$\rho(T) = \text{ActualMinutes}(T) / \text{PlannedMinutes}(T)$	Flagged as 'Needs Revision' if $\rho < 0.50$ and $\Delta \geq 15$ mins (Severity: Medium/High)
Study Deficit	$\Delta_{mins}(T) = \max(0, \text{PlannedMinutes}(T) - \text{ActualMinutes}(T))$	$\Delta \geq 60$m $\rightarrow$ Severity: High | 15m $\leq \Delta <$ 60m $\rightarrow$ Severity: Medium
Missed Task Rule	status = 'missed' $\vee$ (past day uncompleted session)	Severity: High | Tag: 'Missed Task: {topic}' (Mandatory forward recovery)
Priority Weighting	$\exists\, s \in S_T$: priority(s) = 'high' $\wedge$ NeedsRevision	Tag: 'High Priority Exam Target' | Prioritized in adaptive catch-up queue
Future Protection	day_idx(s) > current_day_idx $\wedge$ no_telemetry(s)	Preserved on-track; strictly eliminates false-positive deficits for upcoming days

4. Automated Test Verification (100% Pass Rate Across 53 Tests)

Test Suite	Scope & Validated Invariants	Tests	Pass Rate
Weak Topics Tests	test_weak_topics.py - Missed tasks, sub-50% duration, future protection, deterministic ranking	6	**100%** (6/6)
Study Plan Prompt Tests	test_study_plan_prompt.py - Pydantic validation, AdaptivePlanInput, search_keywords normalization	30	**100%** (30/30)
System Isolation Tests	run_prompt_isolation_tests.py - Obscure electives, zero-hallucination subject name fidelity, milestones	12	**100%** (12/12)
Full Lifecycle E2E Flow	test_e2e_flow.py - Plan → Miss Task → Weak Topic Badge → TF-IDF Coursera Rec → Rebalance	5 Steps	**100%** (5/5)
TOTAL AUTOMATED SUITE	**Comprehensive End-to-End Test Execution**	**53**	**100% PASS**

5. 50% Checkpoint Evaluation Rubric & Faculty Sign-off

Evaluation Parameter	Maximum Marks	Marks Awarded	Evaluator Comments
1. Problem Definition & Hybrid Architecture Design	10	_____ **/ 10**	
2. Deterministic Diagnostics & Zero-Hallucination Guardrails	10	_____ **/ 10**	
3. Coursera TF-IDF Recommendation Pipeline & Retrieval	10	_____ **/ 10**	
4. Full-Stack Implementation & Task Telemetry Dashboard	10	_____ **/ 10**	
5. Test Rigor (53/53 tests passed) & Documentation	10	_____ **/ 10**	
TOTAL MID-TERM EVALUATION MARKS	**50**	_____ **/ 50**	**Result: QUALIFIED**

Project Mentor: _____
Designation: Assistant Professor, Dept. of CSE
Signature: _____
Date: October 31, 2026

Internal Examiner / Coordinator: _____
Designation: Major Project Evaluation Committee
Signature: _____
Date: October 31, 2026